Matrices: Determinants

Questions 1–80

Question 1. It is given that $\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, where a, b, c and d are constants. Write down $|\mathbf{A}|$.

Question 2. In each case, find the determinant of the 2D matrix:

a) $\begin{pmatrix} 3 & 1 \\ 4 & 2 \end{pmatrix}$

b) $\begin{pmatrix} 6 & -4 \\ 3 & 5 \end{pmatrix}$

c) $\begin{pmatrix} 3 & 2 \\ 8 & -5 \end{pmatrix}$

d) $\begin{pmatrix} -6 & -5 \\ 3 & 2 \end{pmatrix}$

Question 3. In each case, find the determinant of the 2D matrix in terms of the constant k, giving your answers in their simplest form:

a) $\begin{pmatrix} k & 5 \\ 2k & 3 \end{pmatrix}$

b) $\begin{pmatrix} k & 4 \\ 5 & 3k \end{pmatrix}$

c) $\begin{pmatrix} 3k & k+1 \\ 4 & k-2 \end{pmatrix}$

d) $\begin{pmatrix} k-2 & k-3 \\ k & 2k+1 \end{pmatrix}$

Question 4. In each case, find the determinant of the 2D matrix in terms of the constants p and q, and simplify as far as possible:

a) $\begin{pmatrix} p & q \\ q & p \end{pmatrix}$

b) $\begin{pmatrix} p & p-3 \\ q & q+2 \end{pmatrix}$

c) $\begin{pmatrix} q-1 & p-1 \\ p+1 & q+1 \end{pmatrix}$

d) $\begin{pmatrix} q+2 & p+3 \\ p-4 & q-3 \end{pmatrix}$

Question 5. Find $\begin{vmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{vmatrix}$, writing your answer in simplified form.

Question 6.

a) Find $\begin{vmatrix} \ln 8 & \ln 2 \\ \ln 4 & \ln 16 \end{vmatrix}$, writing your answer in terms of $\ln 2$.

b) Find $\begin{vmatrix} \ln 81 & \ln 27 \\ \ln 9 & \ln 3 \end{vmatrix}$, writing your answer in terms of $\ln 3$.

Question 7. The triangle T, with vertices at $(0,0)$, $(4,0)$ and $(2,-2)$, is mapped to the triangle T' by the matrix $\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 3 & 5 \end{pmatrix}$.

a) Find the area of T.
b) Find $\det \mathbf{A}$.
c) Find the area of T'.
d) Show how your answers to a) and c) are linked.

Question 8. It is given that $\mathbf{A}$ is a 2×2 matrix such that $\det \mathbf{A} = 6$. The shape P is mapped to the shape Q by the transformation associated with the matrix $\mathbf{A}$. Given that P has an area of 15 units^2, find the area of Q.

Question 9. It is given that $\mathbf{A}$ is a 2×2 matrix such that $\det \mathbf{A} = 3200$, rounded to the nearest 100. The shape C is mapped to the shape D by the transformation associated with the matrix $\mathbf{A}$. Given that C has an area of 25 rounded to the nearest integer, find the smallest possible area of D.

Question 10. The shape S has area 5 cm^2. S is mapped to its image S' under the transformation represented by the matrix $\begin{pmatrix} 5 & 8 \\ 2 & 6 \end{pmatrix}$. Find the area of the shape S'.

Question 11. The triangle T, with vertices at $(-1, 2)$, $(3, 8)$ and $(7, -6)$, is mapped to the triangle T' by the matrix $\mathbf{A} = \begin{pmatrix} 3 & -2 \\ 4 & 6 \end{pmatrix}$.

a) Find the area of T.
b) Find $\det \mathbf{A}$.
c) Hence find the area of T'.

Question 12. The triangle T, with vertices at $(2, 3)$, $(4, -5)$ and $(-1, -1)$, is mapped to the triangle T' by the matrix $\mathbf{A} = \begin{pmatrix} 5-k & 3 \\ -4 & 4 \end{pmatrix}$, where k is a constant. Find, in terms of k:

a) the coordinates of the vertices of T'
b) $\det \mathbf{A}$

Given the area of T' is 320 units^2,

a) find the value of k.

Question 13. The triangle T, with vertices at $(-4, 1)$, $(1, 6)$ and $(2, -7)$, is mapped to the triangle T' by the matrix $\mathbf{A} = \begin{pmatrix} 4-k & 2 \\ -6 & k+3 \end{pmatrix}$, where k is a positive constant. Find, in terms of k:

a) the coordinates of the vertices of T'

b) $\det \mathbf{A}$

Given the area of T' is 140 units^2,

a) find the value of k.

Question 14. The square S, with vertices at $(-3, 2)$, $(4, 2)$, $(4, -5)$ and $(-3, -5)$, is mapped to the rectangle S' by the matrix $\mathbf{A} = \begin{pmatrix} 2k+1 & 5k \\ 3 & k \end{pmatrix}$, where k is a constant. Given that the area of S' is 784 units^2, find the possible values of k.

Question 15. The trapezium T is drawn on a cartesian grid, so that its parallel sides are parallel to the y-axis. The parallel sides have length $\sqrt{2}$ cm and $\sqrt{8}$ cm. The distance between the two parallel sides is 0.25 m. T is mapped to its image T' by the transformation associated with the matrix $\begin{pmatrix} 60 & 0 \\ 0 & \frac{1}{2\sqrt{2}} \end{pmatrix}$. Find the area of the trapezium T'.

Question 16. It is given that $\mathbf{A} = \begin{pmatrix} 6 & 2 \\ 2 & 1 \end{pmatrix}$, $\mathbf{B} = \begin{pmatrix} -2 & 4 \\ 2 & -3 \end{pmatrix}$ and $\mathbf{C} = \begin{pmatrix} 6 & -2 \\ 1 & 3 \end{pmatrix}$. The triangle T, which has area 8 cm^2, is mapped to the triangle T' by the matrix $\mathbf{D}$. Find the area of T' when

a) $\mathbf{D} = \mathbf{A}$
b) $\mathbf{D} = \mathbf{C}$
c) $\mathbf{D} = \mathbf{AB}$
d) $\mathbf{D} = \mathbf{BA}$
e) $\mathbf{D} = \mathbf{AC}$
f) $\mathbf{D} = \mathbf{BC}$
g) $\mathbf{D} = \mathbf{ABC}$

Question 17. Find the value(s) of the constant k in each case:

a) $\begin{vmatrix} k & 3 \\ 5 & 1 \end{vmatrix} = 3$

b) $\begin{vmatrix} k & 8 \\ 5 & 4 \end{vmatrix} = 16$

c) $\begin{vmatrix} k & 3 \\ 4 & k \end{vmatrix} = 4$

d) $\begin{vmatrix} k & 5 \\ k & 9 \end{vmatrix} = 40$

e) $\begin{vmatrix} k & k \\ 3 & 1 \end{vmatrix} = -6$

f) $\begin{vmatrix} k & k \\ k & 10 \end{vmatrix} = 9$

g) $\begin{vmatrix} 2k & k \\ 4 & k \end{vmatrix} = 2$

h) $\begin{vmatrix} 3k & 6 \\ k & 4k \end{vmatrix} = 6$

i) $\begin{vmatrix} k+1 & 3 \\ 4 & k \end{vmatrix} = 8$

j) $\begin{vmatrix} 2k-1 & 4 \\ 5 & k-2 \end{vmatrix} = 7$

k) $\begin{vmatrix} k & k-4 \\ 2k+5 & 6 \end{vmatrix} = -6$

l) $\begin{vmatrix} 3 & k+1 \\ 8-k & 7k \end{vmatrix} = 15$

Question 18. Given $\mathbf{A} = \begin{pmatrix} k & 2 \\ 3 & 8 \end{pmatrix}$, find the values of constant k for which $\mathbf{A}$ has a positive determinant.

Question 19. Given $\mathbf{A} = \begin{pmatrix} 3k & -4 \\ 2 & -6 \end{pmatrix}$, find the values of constant k for which $\mathbf{A}$ has a negative determinant.

Question 20. Given $\mathbf{A} = \begin{pmatrix} k & k+2 \\ 2 & 4k-4 \end{pmatrix}$, find the values of constant k for which $\mathbf{A}$ has a positive determinant.

Question 21. Given $\mathbf{A} = \begin{pmatrix} k-1 & 1-k \\ k & 2k+1 \end{pmatrix}$, find the values of constant k for which $\mathbf{A}$ has a negative determinant.

Question 22. Given $\mathbf{A} = \begin{pmatrix} 2+i & 3 \\ -2 & -i \end{pmatrix}$, show that $\det \mathbf{A}$ can be written in the form $a + bi$, where a and b are integers.

Question 23. In each case, considering the transformation as a matrix, determine whether the determinant of the matrix would be positive or negative:

a) Rotation.
b) Reflection.
c) Enlargement.

Question 24. The transformation M has associated matrix $\mathbf{A}$ where $\det \mathbf{A} = k$. State the possible values of k in each case:

a) M preserves area.
b) M preserves orientation.

Question 25. In each case, by only considering the determinant, determine whether the matrix represents a reflection.

a) $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

b) $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$

c) $\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$

d) $\begin{pmatrix} 0 & -1 \\ 0 & 1 \end{pmatrix}$

e) $\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$

f) $\begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$

Question 26. In each case, by only considering the determinant, determine whether the matrix reverses orientation.

a) $\begin{pmatrix} 4 & -6 \\ -9 & 3 \end{pmatrix}$

b) $\begin{pmatrix} -5 & 7 \\ 1 & -2 \end{pmatrix}$

c) $\begin{pmatrix} 8 & -9 \\ 10 & -3 \end{pmatrix}$

d) $\begin{pmatrix} 21 & 10 \\ 15 & 7 \end{pmatrix}$

Question 27. The vertices of a square with area 18 units^2, read clockwise from A, are A, B, C and D. The square is mapped to its image by the transformation represented by the matrix M, where $\det M = -10$, so that A is mapped to A', B is mapped to B', etc.

a) Write down the area of the square's image.

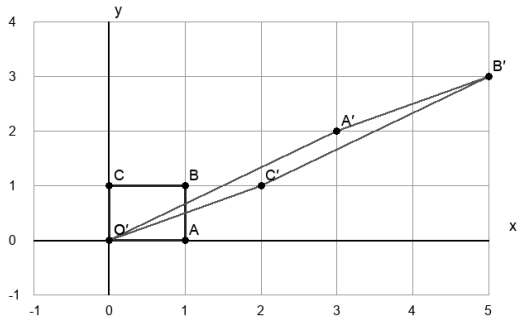
b) Write down the vertices of the square's image, read clockwise from A'.

Question 28. The triangle T is mapped to its image T' by a transformation represented by the matrix A. Given that $\det A = 4k^2 + 2$, where k is a real number, explain why the transformation will not reverse the orientation of the vertices of T.

Question 29. The triangle T is mapped to its image T' by a transformation represented by the matrix A. Given that $\det A = 9 - 4k^2$, where k is a real number, determine the values of k for which the transformation will reverse the orientation of T.

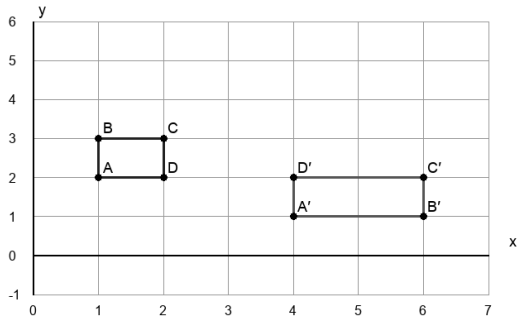
Question 30. The shape S is mapped to its image S' by a transformation represented by the matrix $A = \begin{pmatrix} 80 & 3k \\ 9 & 2 \end{pmatrix}$, where k is a positive integer. Find the smallest value of k such that the transformation reverses orientation.

Question 31. It is given that $A = \begin{pmatrix} 3 & 2 \\ 2 & 1 \end{pmatrix}$. The unit square S is mapped to its image S' under the transformation represented by A, as shown in the diagram below.



a) Write down the area of S'.
b) Describe how the matrix A has affected the order of the vertices of shape S during the transformation.

Question 32. It is given that $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. The square $ABCD$ is mapped to its image $A'B'C'D'$ under the transformation represented by the matrix A, as shown in the diagram below.



a) Write down $\det A$.
b) Find the values of a, b, c and d.

Question 33. It is given that $A = \begin{pmatrix} \frac{3}{5} & \frac{3\sqrt{3}}{5} \\ \frac{3\sqrt{3}}{5} & -\frac{3}{5} \end{pmatrix}$ is a combination of an enlargement about the origin and a reflection. By considering $\det A$, determine the scale factor of the enlargement.

Question 34. It is given that $A = \begin{pmatrix} 4\sqrt{3} & -4 \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{pmatrix}$ is a combination of a stretch parallel to the x-axis and an anticlockwise rotation about the origin.

a) By considering $\det A$, determine the scale factor of the stretch.

b) Fully describe the rotation.

Question 35. It is given that $A = \begin{pmatrix} k-2 & k-3 \\ k+3 & k+2 \end{pmatrix}$, where k is a constant.

a) Show that $\det A > 0$ for all values of k.

b) State the significance of $\det A > 0$.

Question 36. It is given that $A = \begin{pmatrix} 8 & -5 \\ k & -2 \end{pmatrix}$ and $B = \begin{pmatrix} 9 & 2 \\ 13 & k \end{pmatrix}$, where k is a constant. The transformation represented by AB preserves area but reverses orientation. Find the possible values of k.

Question 37. It is given that $A = \begin{pmatrix} 2 & -4 \\ 3 & 6 \end{pmatrix}$ and $B = \begin{pmatrix} -1 & 2 \\ -8 & 4 \end{pmatrix}$.

a) Find $\det A$.
b) Find $\det B$.
c) Find the matrix AB.
d) Find $\det AB$.
e) Explain how your answers to a), b) and d) relate to each other.

Question 38. Matrices A and B are square matrices of the same order, such that $\det A = 5$ and $\det B = -6$.

a) Find $\det AB$.
b) Find $\det BA$.
c) Find $\det ABA$.
d) Find $\det BABABA$.

Question 39. Matrices A and B are square matrices of the same order, such that $\det AB = 20$ and $\det B = 5$. Find $\det A^5$.

Question 40. Matrices A and B are square matrices of the same order, such that $AB = I$. Describe the relationship between $\det A$ and $\det B$.

Question 41. It is given that $A = \begin{pmatrix} 4 & k \\ 6 & 2 \end{pmatrix}$, where k is a constant. Find the possible values of k given that $|A^2| = 16$.

Question 42. It is given that $A = \begin{pmatrix} 5 & 2 \\ -4 & k \end{pmatrix}$, where k is a real constant. Find the value of k given that $|A^3| = 125$.

Question 43. For a constant k, it is given that $A = \begin{pmatrix} 4 & 3+i \\ 2 & -i \end{pmatrix}$ and $B = \begin{pmatrix} k+1 & 3 \\ 2-i & i \end{pmatrix}$. Find the value of k such that AB has argument $\frac{\pi}{4}$.

Question 44. The shape S is mapped to the shape S' by the transformation represented by the matrix A, where $A = \begin{pmatrix} 2x-3 & x-5 \\ x+6 & x-2 \end{pmatrix}$. S has an area of 30 cm^2. Given that x can vary, find the minimum area of the shape S'.

Question 45. The shape S is mapped to the shape S' by the transformation represented by the matrix A, where $A = \begin{pmatrix} 2x^2+1 & 4x-3 \\ 4x+3 & x^2+3 \end{pmatrix}$. S has an area of 16 cm^2. Given that x can vary, find the minimum area of the shape S'.

Question 46. For constant k, it is given that $A = \begin{pmatrix} k+1 & k-3 \\ k-4 & k+2 \end{pmatrix}$ and $B = \begin{pmatrix} 2k-1 & 3k+1 \\ 1-3k & 2k-1 \end{pmatrix}$. Find the values of k such that $\det AB = 0$.

Question 47. For constant k, it is given that $A = \begin{pmatrix} 2k+3 & k+1 \\ k-2 & k-4 \end{pmatrix}$ and $B = \begin{pmatrix} 3k+1 & k-3 \\ k+4 & k+2 \end{pmatrix}$. Find the values of k such that $\det A = \det B$.

Question 48.

a) (i) Prove that $\begin{vmatrix} ka & b \\ kc & d \end{vmatrix} = k \begin{vmatrix} a & b \\ c & d \end{vmatrix}$.

(ii) Prove that $\begin{vmatrix} a & kb \\ c & kd \end{vmatrix} = k \begin{vmatrix} a & b \\ c & d \end{vmatrix}$.

(iii) Prove that $\begin{vmatrix} ka & kb \\ c & d \end{vmatrix} = k \begin{vmatrix} a & b \\ c & d \end{vmatrix}$.

(iv) Prove that $\begin{vmatrix} ka & kb \\ kc & kd \end{vmatrix} = k^2 \begin{vmatrix} a & b \\ c & d \end{vmatrix}$.

b) It is given that $\begin{vmatrix} 3 & 4 \\ -2 & 5 \end{vmatrix} = 23$. Using part a), and without calculating the determinant directly, write down:

(i) $\begin{vmatrix} 6 & 8 \\ -2 & 5 \end{vmatrix}$

(ii) $\begin{vmatrix} 3 & 12 \\ -2 & 15 \end{vmatrix}$

(iii) $\begin{vmatrix} 30 & 4 \\ -20 & 5 \end{vmatrix}$

(iv) $\begin{vmatrix} 15 & 20 \\ -20 & 50 \end{vmatrix}$

Question 49. In each case, find the determinant of the 3D matrix:

a) $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 6 \end{pmatrix}$

b) $\begin{pmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 6 \end{pmatrix}$

c) $\begin{pmatrix} 3 & 0 & 0 \\ 0 & 2 & 3 \\ 0 & 1 & 6 \end{pmatrix}$

d) $\begin{pmatrix} 3 & 0 & 1 \\ 0 & 2 & 3 \\ 0 & 1 & 6 \end{pmatrix}$

e) $\begin{pmatrix} 3 & 2 & 1 \\ 2 & 2 & 3 \\ 1 & 1 & 6 \end{pmatrix}$

Question 50. The cuboid C has a volume of 33 cm^3. C is mapped to the object C' by the matrix $A = \begin{pmatrix} 1 & 1 & 2 \\ 3 & 1 & 1 \\ 1 & 1 & 4 \end{pmatrix}$.

Find the volume of C'.

Question 51. The cuboid C, with vertices $(0,0,0)$, $(5,0,0)$, $(5,6,0)$, $(0,6,0)$, $(0,0,8)$, $(5,0,8)$, $(5,6,8)$ and $(0,6,8)$, is mapped to the shape C' by the matrix $A = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & \frac{1}{12} \end{pmatrix}$.

a) Find the volume of C.
b) Find $\det A$.
c) Find the volume of C'.

Question 52. The shape S has volume 9 cm^3. S is mapped to its image S' under the transformation represented by the matrix $\begin{pmatrix} 2 & -3 & 4 \\ 1 & 0 & 6 \\ -3 & 2 & 4 \end{pmatrix}$. Find the volume of the shape S'.

Question 53. It is given that $A = \begin{pmatrix} 3 & 2 & 0 \\ 1 & 2 & -1 \\ 3 & 3 & -1 \end{pmatrix}$.

a) Find $\det A$.

b) Given that the matrix A represents a linear transformation, state two geometrical consequences of your answer to a).

Question 54. In each case, find the volume scale factor of the transformation associated with each of the following matrices:

a) $\begin{pmatrix} 3 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 1/9 \end{pmatrix}$

b) $\begin{pmatrix} -2 & 0 & 0 \\ 0 & 18 & 0 \\ 0 & 0 & 3 \end{pmatrix}$

c) $\begin{pmatrix} 5 & 1 & 7 \\ 2 & 0 & 0 \\ 5 & 0 & 1 \end{pmatrix}$

d) $\begin{pmatrix} 3 & -2 & -4 \\ 2 & 4 & -3 \\ 3 & 1 & 3 \end{pmatrix}$

Question 55. The equation of a sphere centred at (a, b, c) with radius r is $(x-a)^2+(y-b)^2+(z-c)^2=r^2$. The sphere S has equation $x^2+y^2+z^2=27$. S is mapped to its image S' by the transformation associated with the matrix $\begin{pmatrix} 1/3 & 0 & 0 \\ 0 & 1/\sqrt{3} & 0 \\ 0 & 0 & 1/\sqrt{2} \end{pmatrix}$. Find the exact volume of the sphere S'.

Question 56. Given $A = \begin{pmatrix} 2 & 3 & 8 \\ -4 & 3 & 1 \\ -3 & 2 & 1 \end{pmatrix}$, show that $\det(A^2) = (\det A)^2$.

Question 57. Given $A = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$, show that $\det(A^3) = (\det A)^3$.

Question 58. It is given that $A = \begin{pmatrix} 0 & 2 & 0 \\ 3 & 0 & -5 \\ 0 & 4 & 0 \end{pmatrix}$, $B = \begin{pmatrix} 5 & 0 & 1 \\ 0 & 2 & 0 \\ 3 & 0 & 4 \end{pmatrix}$ and $C = \begin{pmatrix} -1 & 2 & 1 \\ 3 & -2 & 4 \\ -9 & 1 & 4 \end{pmatrix}$. The shape S, which has volume 12 cm^3, is mapped to the shape S' by the matrix D. Find the volume of S' when

a) $D = A$
b) $D = B$
c) $D = C$
d) $D = AB$
e) $D = AC$
f) $D = BC$
g) $D = ABC$

Question 59. In each case, find the determinant of the 3D matrix in terms of the constant k:

a) $\begin{pmatrix} k & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & k \end{pmatrix}$

b) $\begin{pmatrix} k & 0 & 1 \\ 0 & k & 0 \\ 4 & 0 & k \end{pmatrix}$

c) $\begin{pmatrix} k & 1 & 0 \\ 1 & 2 & 3 \\ 2 & 1 & 4 \end{pmatrix}$

d) $\begin{pmatrix} k & 0 & k \\ 3 & 5 & -1 \\ 2 & -3 & 2 \end{pmatrix}$

e) $\begin{pmatrix} 1 & 2 & 4 \\ k & 2 & 1 \\ -4 & 3 & 5 \end{pmatrix}$

f) $\begin{pmatrix} 4 & k & k \\ k & 3 & 2 \\ k & 2 & 3 \end{pmatrix}$

g) $\begin{pmatrix} 5 & 3k & 3 \\ -2 & 1 & 4k \\ 2k & 3 & -5 \end{pmatrix}$

Question 60. Find the exact real value(s) of the constant k in each case:

a) $\begin{vmatrix} k & 1 & 0 \\ 2 & 4 & 3 \\ -1 & 2 & 3 \end{vmatrix} = 1$

b) $\begin{vmatrix} 3 & k & 4 \\ -1 & 2 & -5 \\ 3 & 2 & 7 \end{vmatrix} = 8$

c) $\begin{vmatrix} 2 & 3 & -5 \\ -2 & k & 2 \\ 1 & -4 & 1 \end{vmatrix} = -33$

d) $\begin{vmatrix} k & 2 & 5 \\ 9 & 8 & 2 \\ k & 3 & 1 \end{vmatrix} = 49$

e) $\begin{vmatrix} k & 0 & 0 \\ 0 & k & 0 \\ 0 & 0 & k+1 \end{vmatrix} = 150$

f) $\begin{vmatrix} 2k & 1 & 3k \\ 1 & 5 & 1 \\ 3k & 1 & 2k \end{vmatrix} = -23$

g) $\begin{vmatrix} k+1 & 2 & 2k \\ 1 & k & 3 \\ k & 0 & k \end{vmatrix} = -2$

Question 61. It is given that $A = \begin{pmatrix} 0 & k & 2 \\ -23 & k+1 & k \\ k-3 & 1 & \frac{1}{2} \end{pmatrix}$, where k is either a real or purely imaginary constant.

a) When $k = 5$, $\det A = p$. Find the value of p.

b) Find all the possible values of k when $\det A = p$.

Question 62. It is given that $A = \begin{pmatrix} k-2 & 2k & 3k+3 \\ 2k-4 & 4k & k+1 \\ 2-k & 8k & 9k+9 \end{pmatrix}$, where k is a constant. Write $\det A$ in fully factorised form.

Question 63. Explain what is meant by a singular matrix.

Question 64. Determine which of the following matrices are singular:

a) $\begin{pmatrix} 4 & 2 \\ 4 & 2 \end{pmatrix}$

b) $\begin{pmatrix} 3 & 18 \\ 1 & 6 \end{pmatrix}$

c) $\begin{pmatrix} 1 & 6 \\ -2 & -5 \end{pmatrix}$

d) $\begin{pmatrix} 6 & -6 \\ -8 & -8 \end{pmatrix}$

e) $\begin{pmatrix} \sqrt{3} & -\sqrt{6} \\ -\sqrt{6} & 2\sqrt{3} \end{pmatrix}$

f) $\begin{pmatrix} k+3 & k \\ 2k+6 & 2k \end{pmatrix}$, where k is a constant.

Question 65. It is given that $A = \begin{pmatrix} 4 & 3 \\ 8 & 6 \end{pmatrix}$. The shape S is mapped to its image S' under the transformation represented by the matrix A. It is also given that shape S has area 16 cm^2.

a) Find the area of shape S'.

b) Every point in shape S is mapped to part of a line with equation $y = kx$. Find the value of k.

Question 66. Each of the following matrices are singular. Find the value(s) of the constant k in each case:

a) $\begin{pmatrix} k-3 & 2k+9 \\ 8 & -4 \end{pmatrix}$

b) $\begin{pmatrix} k-1 & k+5 \\ k+2 & k+3 \end{pmatrix}$

c) $\begin{pmatrix} k+6 & k-4 \\ 3-k & 8-k \end{pmatrix}$

d) $\begin{pmatrix} 2k+3 & k+1 \\ k+2 & k+2 \end{pmatrix}$

e) $\begin{pmatrix} 3k-2 & k+4 \\ k-2 & k+3 \end{pmatrix}$

Question 67. It is given that $A = \begin{pmatrix} 3 & p \\ -2 & 4 \end{pmatrix}$ and $B = \begin{pmatrix} 4 & 2 \\ 6 & q \end{pmatrix}$, where p and q are constants. Show that AB is singular when $pq + 6q = 3p + 18$.

Question 68. Determine which of the following 2D matrices are singular:

a) $M = \begin{pmatrix} 4 & 30 \\ \frac{1}{3} & 5/2 \end{pmatrix}$

b) $N = \begin{pmatrix} 10 & 2/3 \\ -155 & -1/3 \end{pmatrix}$

c) MN

d) NM

Question 69. Determine which of the following 3D matrices are singular:

a) $\begin{pmatrix} 6 & 3 & 4 \\ 3 & 0 & 1 \\ 2 & -1 & 0 \end{pmatrix}$

b) $\begin{pmatrix} 0 & -6 & 4 \\ -1 & 1 & 0 \\ 0 & 3 & -2 \end{pmatrix}$

c) $\begin{pmatrix} -1 & 2 & 3 \\ -2 & 3 & -5 \\ 1 & -2 & -2 \end{pmatrix}$

d) $\begin{pmatrix} -\sqrt{2} & 2\sqrt{2} & 2\sqrt{2} \\ -\sqrt{2} & 2\sqrt{8} & 1 \\ \sqrt{2} & -2\sqrt{2} & -\sqrt{8} \end{pmatrix}$

Question 70. Determine which of the following 3D matrices are singular:

a) $A = \begin{pmatrix} 1 & 1 & 2 \\ 3 & 2 & 1 \\ 5 & 4 & -1 \end{pmatrix}$

b) $B = \begin{pmatrix} -2 & 3 & 1 \\ 4 & 1 & 5 \\ 3 & -2 & 1 \end{pmatrix}$

c) AB

d) BA

Question 71. It is given that $A = \begin{pmatrix} 2 & k & 3 \\ 2 & 2 & k \\ 0 & 1 & -1 \end{pmatrix}$, where k is a constant. Show that A is never singular.

Question 72. It is given that $A = \begin{pmatrix} k^2 & 1 & 2 \\ -k^2 & 2k^2 & 3 \\ -1 & 1 & -2 \end{pmatrix}$, where k is a constant. Show that A is never singular.

Question 73. It is given that $A = \begin{pmatrix} -\sin\theta & 1 & \cos\theta \\ 0 & 1 & 0 \\ \cos\theta & 1 & \sin\theta \end{pmatrix}$. Show that A is never singular.

Question 74. It is given that $A = \begin{pmatrix} \sin\theta - 1 & 1 & 1 - \cos\theta \\ 0 & 1 & 0 \\ 1 - \cos\theta & 1 & 1 - \sin\theta \end{pmatrix}$. Show that A is never singular.

Question 75. Each of the following matrices are singular. Find the value(s) of the constant k in each case:

a) $\begin{pmatrix} 7 & -1 & k \\ 2 & 1 & k \\ 0 & 5 & 4 \end{pmatrix}$

b) $\begin{pmatrix} -2 & k & 4 \\ -3 & 1 & k \\ 1 & 2 & 4 \end{pmatrix}$

c) $\begin{pmatrix} k+2 & 2 & -3 \\ 2 & 2 & 3 \\ 1 & 0 & k-2 \end{pmatrix}$

Question 76. It is given that $A = \begin{pmatrix} 2k & 1 & k \\ 4k & 5 & 3 \\ 3 & -6 & k \end{pmatrix}$. The square-based pyramid S has vertices A, B, C, D and E, and has a volume of 20 cm^3. S is mapped to its image S', which has vertices A', B', C', D', E', under the transformation represented by the matrix A. Find the values of k such that the points A', B', C', D', E' are coplanar.

Question 77. Harry says that if at least one of two square matrices are singular, then their product will be a singular matrix. Explain why Harry is correct.

Question 78. It is given that A and B are non-singular 2D matrices. Prove that in general $\det(A + B) \neq \det A + \det B$.

Question 79. Find an example of non-zero, non-singular 2D matrices A and B such that $\det(A+B) = \det A + \det B$.

Question 80. It is given that A and B are non-singular 2D matrices. Prove that $\det AB = \det A \times \det B$.